Register [DaveW Ditroni(X]	Register ID	Description	Area	Function	IPEM Plus E36 Hex	E36 Manf Defaults	E32	E36	HA/ESPHome Platform Phase	HA/ESPHome Platform	HA/ESPHome Default atm90e32	HA/ESPHome Platform atm90e36	HA/ESPHome Name	Comments				Variables
0x00	SoftReset	Software Reset	Status Registers E36		**0x0000**	0x0000	No	Yes						Force Soft Reset > Write 0x789A. Wait 6ms				
0x01	SysStatus0	System Status0	Status Registers E36	GetSysStatus0	**0x0000**	x0210	No	Yes						System Status 0. S0:0xC0				
0x02	SysStatus1	System Status1	Status Registers E36	GetSysStatus1	**0x0000**	0x0000	No	Yes						System Status 1. S1:0x0 or 0xF or 0x400				
0x03	FuncEn0	Function Enable0	Status Registers E36		**0x0000**		No	Yes						Function Enable 0				
0x04	FuncEn1	Function Enable1	Status Registers E36		**0x0000**		No	Yes						Function Enable 1				
0x07	ZXConfig	Zero-Crossing Config	Status Registers		**0x0001**	0x0001	Yes	Yes										
0x08	SagTh	Voltage Sag Th	Status Registers	vSagTh	**0x0001**	0x0000	Yes	Yes						SagTh = ([90 or 190] * 100 * sqrt(2)) / (2 * _uGainCT1 / 32768)				
0x09	PhaseLossTh	Voltage Phase Losing Th	Status Registers		**0x0000**	0x0000	Yes	Yes										
0x0A	INWarnTh0	N Current Line Th	Status Registers		**0xFFFF**	0xFFFF	Yes	Yes										
0x0A	INWarnTh	Neutral Current (Calculated) Warning Threshold	Status Registers E36				No	Yes										
0x0B	INWarnTh1	Voltage ADC Th	Status Registers E36		**0xFFFF**	0xFFFF	No	Yes										
0x0C	THDNUTh	Voltage THD Th	Status Registers E36		**0xFFFF**	0xFFFF	No	Yes										
0x0D	THDNITh	Current THD Th	Status Registers E36		**0xFFFF**	0xFFFF	No	Yes										
0x0E	DMACtrl	DMA Int. Control	Status Registers E36	FreqLoThresh	**0x7E44**	0x7E44	No	Yes										
0x0F	LastSPIData	Last Read/Write SPI Value	Status Registers E36		**0x7E44**	0x7E44	No	Yes										
0x10	DetectCtrl	Current Detect Controller	Low Power Registers		**x**	x	Yes	Yes										
0x11	DetectTh1	Channel CT1 Currrent Threshold in Detection Mode	Low Power Registers		**x**	x	Yes	Yes	phase_a									
0x12	DetectTh2	Channel CT2 Currrent Threshold in Detection Mode	Low Power Registers		**x**	x	Yes	Yes	phase_b									
0x13	DetectTh3	Channel CT3 Currrent Threshold in Detection Mode	Low Power Registers		**x**	x	Yes	Yes	phase_c									
0x14	PMoffsetCT1	Ioffset for Phase A in Partial Measurment Mode	Low Power Registers		**x**	x	Yes	Yes	phase_a									
0x15	PMoffsetCT2	Ioffset for Phase B vin Partial Measurment Mode	Low Power Registers		**x**	x	Yes	Yes	phase_b									
0x16	PMoffsetCT3	Ioffset for Phase C in Partial Measurment Mode	Low Power Registers		**x**	x	Yes	Yes	phase_c									
0x17	PMPGA	OGAgain Configuration in in Partial Measurment Mode	Low Power Registers		**x**	x	Yes	Yes										
0x18	PMIRMSCT1	irms for Phase A in Partial Measurment Mode	Low Power Registers		**x**	x	Yes	Yes	phase_a									
0x19	PMIRMSCT2	irms fDetectTh3r Phase B in Partial Measurment Mode	Low Power Registers		**x**	x	Yes	Yes	phase_b									
0x1A	PMIRMSCT3	irms for Phase C in Partial Measurment Mode	Low Power Registers		**x**	x	Yes	Yes	phase_c									
0x1B	PMConfig	Measure Configuraiton in Partial Measurment Mode	Low Power Registers		**x**	x	Yes	Yes										
0x1C	PMAvgSamples	Number of 8K samples to be averages in RMS/mean computation	Low Power Registers		**x**	x	Yes	Yes										
0x1D	PMIRMSLSB	LSB bits of PMRRrms (A/B/C)	Low Power Registers		**x**	x	Yes	Yes										
0x30	ConfigStart	Configuration Start	Configuration Registers E36		**0x5678**	0x6886	No	Yes										
0x31	PLconstH	High Word of PL_Constant	Configuration Registers		**0x861**	0x0861	Yes	Yes										
0x32	PLconstL	Low Word of PL_Constant	Configuration Registers		**0xC468**	0xC468	Yes	Yes						PL Constant LSB (default) - this is 4C68 in the application note, which is incorrect				
0x33	MMode0	Metering Mode Config. Mains Frequency	Configuration Registers	lineFreq	**0x185**	0x0087	Yes	Yes	line_frequency					0x1087. Mode Config (60 Hz, 3P4W). PL Constant MSB (default) - Meter Constant = 3200 - PL Constant = 140625000 [E32 0x1087)				
0x34	MMode1	Metering Mode Config. PGA Gain Configuration for Current Channels	Configuration Registers	pgagain	**0x5555**	0x0000	Yes	Yes	gain_pga					0x1500. 0x5555 (x2) // 0x0000 (1x)				
0x35	PStartTh	Startup Power Th (P)	Configuration Registers		**0x0000**	0x0000	Yes	Yes						Active Startup Power Threshold [E32 All phase Active Startup Power Threshold - 50% of startup current = 0.02A/0.00032 = 7500]				
0x36	QStartTh	Startup Power Th (Q)	Configuration Registers		**0x0000**	0x0000	Yes	Yes						Reactive Startup Power Threshold [E32 All phase Active Startup Power Threshold - 50% of startup current = 0.02A/0.00032 = 7500]				
0x37	SStartTh	Startup Power Th (S)	Configuration Registers		**0x0000**	0x0000	Yes	Yes						Apparent Startup Power Threshold [E32 All phase Active Startup Power Threshold - 50% of startup current = 0.02A/0.00032 = 7500]				
0x38	PPhaseTh	Startup Power Accum Th (P)	Configuration Registers		**0x0000**	0x0000	Yes	Yes						Active Phase Threshold [E32 Each phase Active Phase Threshold = 10% of startup current = 0.002A/0.00032 = 750]				
0x39	QPhaseTh	Startup Power Accum Th (Q)	Configuration Registers		**0x0000**	0x0000	Yes	Yes						Reactive Phase Threshold [E32 Each phase Active Phase Threshold = 10% of startup current = 0.002A/0.00032 = 750]				
0x3A	SPhaseTh	Startup Power Accum Th (S)	Configuration Registers		**0x0000**	0x0000	Yes	Yes						Apparent Phase Threshold [E32 Each phase Active Phase Threshold = 10% of startup current = 0.002A/0.00032 = 750]				
0x3B	CSZero	Checksum 0	Configuration Registers E36		**0x41C5**	0x421C	No	Yes						Checksum 0				
0x40	CalStart	Cal Start	Calibration Registers E36		**0x5678**	0x6886	No	Yes						Metering calibration startup				
0x41	PoffsetCT1	CT1 Line Power Offset (P)	Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_a					CT1 line active power offset [FFDC]				
0x42	QoffsetCT1	CT1 Line Power Offset (Q)	Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_a					CT1 line reactive power offset				
0x43	PoffsetCT2	CT2 Line Power Offset (P)	Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_b					CT2 line active power offset				
0x44	QoffsetCT2	CT2 Line Power Offset (Q)	Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_b					CT2 line reactive power offset				
0x45	PoffsetCT3	CT3 Line Power Offset (P)	Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_c					CT3 line active power offset				
0x46	QoffsetCT3	CT3 Line Power Offset (Q)	Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_c					CT3 line reactive power offset				
0x47	PQGainCT1	CT1 Line Calibration Gain	Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_a					CT1 Line calibration gain				
0x48	PhiCT1	CT1 Line Calibration Angle	Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_a					CT1 Line calibration angle				
0x49	PQGainCT2	CT2 Line Calibration Gain	Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_b					CT2 Line calibration gain				
0x4A	PhiCT2	CT2 Line Calibration Angle	Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_b					CT3 Line calibration angle				
0x4B	PQGainCT3	CT3 Line Calibration Gain	Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_c					CT3 Line calibration gain				
0x4C	PhiCT3	CT3 Line Calibration Angle	Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_c					CT3 Line calibration angle				
0x4D	CSOne	Checksum 1	Calibration Registers E36		**0x0000**	0x0000	No	Yes						Checksum 1				
0x50	HaRMStart	Harmonic Cal Start	Fundamental Harmonic Energy Calibration Registers E36		**0x5678**	0x6886	No	Yes						Harmonic Power				
0x51	PoffsetCT1F	CT1 Fund Power Offset (P)	Fundamental Harmonic Energy Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_a									
0x52	PoffsetCT2F	CT2 Fund Power Offset (P)	Fundamental Harmonic Energy Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_b									
0x53	PoffsetCT3F	CT3 Fund Power Offset (P)	Fundamental Harmonic Energy Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_c									
0x54	PGainCT1F	CT1 Fund Power Gain (P)	Fundamental Harmonic Energy Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_a									
0x55	PGainCT2F	CT2 Fund Power Gain (P)	Fundamental Harmonic Energy Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_b									
0x56	PGainCT3F	CT3 Fund Power Gain (P)	Fundamental Harmonic Energy Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_c									
0x57	CSTwo	Checksum 2	Fundamental Harmonic Energy Calibration Registers E36		**0x0000**	0x0000	No	Yes						Checksum 2				
0x60	AdjStart	Measurement Cal Start	Measurement Calibration Registers E36		**0x5678**	0x6886	No	Yes						Measurement calibration [AdjStart] Alternative Value to Test 0xFD7F				
0x61	UGainCT1	CT1 Voltage RMS Gain	Measurement Calibration Registers	_uGainCT1	**0x4EE8**	0xCE40	Yes	Yes	phase_a	gain_voltage	7305			Base value for 240V is 20200. 120V is 9700. ~100 per 1.0 V RMS. 12V setting @ ~19V RMS				
0x62	IGainCT1	CT1 Current RMS Gain	Measurement Calibration Registers	_iGainCT1	**0x82DC**	0x7530	Yes	Yes	phase_a	gain_ct	27961							
0x63	UoffsetCT1	CT1 Voltage Offset	Measurement Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_a									
0x64	IoffsetCT1	CT1 Current Offset	Measurement Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_a									
0x65	UGainCT2	CT2 Voltage RMS Gain	Measurement Calibration Registers	_uGainCT2	**0x4EE8**	0xCE40	Yes	Yes	phase_b	gain_voltage	7305			Base value for 240V is 20200. 120V is 9700. ~100 per 1.0 V RMS. 12V setting @ ~19V RMS				
0x66	IGainCT2	CT2 Current RMS Gain	Measurement Calibration Registers	_iGainCT2	**0x82DC**	0x7530	Yes	Yes	phase_b	gain_ct	27961							
0x67	UoffsetCT2	CT2 Voltage Offset	Measurement Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_b									
0x68	IoffsetCT2	CT2 Current Offset	Measurement Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_b									
0x69	UGainCT3	CT3 Voltage RMS Gain	Measurement Calibration Registers	_uGainCT3	**0x4EE8**	0xCE40	Yes	Yes	phase_c	gain_voltage	7305			Base value for 240V is 20200. 120V is 9700. ~100 per 1.0 V RMS. 12V setting @ ~19V RMS				
0x6A	IGainCT3	CT3 Current RMS Gain	Measurement Calibration Registers	_iGainCT3	**0x82DC**	0x7530	Yes	Yes	phase_c	gain_ct	27961							
0x6B	UoffsetCT3	CT3 Voltage Offset	Measurement Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_c									
0x6C	IoffsetCT3	CT3 Current Offset	Measurement Calibration Registers		**0x0000**	0x0000	Yes	Yes	phase_c									
0x6D	IgainN	N Current Gain	Measurement Calibration Registers E36		**0xFD7F**	0x7530	No	Yes	phase_n									
0x6E	IoffsetN	N Current Offset	Measurement Calibration Registers E36		**0x0000**	0x0000	No	Yes	phase_n									
0x6F	CSThree	Checksum 3	Measurement Calibration Registers E36		**0x7A38**	0x8EBE	No	Yes						Checksum 3				
0x80	APenergyT	Total Forward Active	Energy Registers	GetImportEnergy	**x**	x	Yes	Yes						These registers accumulate energy and are cleared after being read				
0x81	APenergyCT1	CT1 Forward Active	Energy Registers		**x**	x	Yes	Yes	phase_a									
0x82	APenergyCT2	CT2 Forward Active	Energy Registers		**x**	x	Yes	Yes	phase_b									
0x83	APenergyCT3	CT3 Forward Active	Energy Registers		**x**	x	Yes	Yes	phase_c									
0x84	ANenergyT	Total Reverse Active	Energy Registers	GetExportEnergy	**x**	x	Yes	Yes						Reverse (Export) Active Energy				
0x85	ANenergyCT1	CT1 Reverse Active	Energy Registers	GetExportEnergy	**x**	x	Yes	Yes	phase_a					Reverse (Export) Active Energy				
0x86	ANenergyCT2	CT2 Reverse Active	Energy Registers	GetExportEnergy	**x**	x	Yes	Yes	phase_b					Reverse (Export) Active Energy				
0x87	ANenergyCT3	CT3 Reverse Active	Energy Registers	GetExportEnergy	**x**	x	Yes	Yes	phase_c					Reverse (Export) Active Energy				
0x88	RPenergyT	Total Forward Reactive	Energy Registers	GetImportReactiveEnergy	**x**	x	Yes	Yes						Forward Reactive Energy				
0x89	RPenergyCT1	CT1 Forward Reactive	Energy Registers	GetImportReactiveEnergy	**x**	x	Yes	Yes	phase_a					Forward Reactive Energy				
0x8A	RPenergyCT2	CT2 Forward Reactive	Energy Registers	GetImportReactiveEnergy	**x**	x	Yes	Yes	phase_b					Forward Reactive Energy				
0x8B	RPenergyCT3	CT3 Forward Reactive	Energy Registers	GetImportReactiveEnergy	**x**	x	Yes	Yes	phase_c					Forward Reactive Energy				
0x8C	RNenergyT	Total Reverse Reactive	Energy Registers	GetExportReactiveEnergy	**x**	x	Yes	Yes						Reverse ReActive Energy				
0x8D	RNenergyCT1	CT1 Reverse Reactive	Energy Registers	GetExportReactiveEnergy	**x**	x	Yes	Yes	phase_a					Reverse ReActive Energy				
0x8E	RNenergyCT2	CT2 Reverse Reactive	Energy Registers	GetExportReactiveEnergy	**x**	x	Yes	Yes	phase_b					Reverse ReActive Energy				
0x8F	RNenergyCT3	CT3 Reverse Reactive	Energy Registers	GetExportReactiveEnergy	**x**	x	Yes	Yes	phase_c					Reverse ReActive Energy				
0x90	SAenergyT	Total Apparent Energy	Energy Registers	GetImportApparentEnergy	**x**	x	Yes	Yes						Import Apparent Energy				
0x91	SenergyCT1	CT1 Apparent Energy	Energy Registers	GetImportApparentEnergy	**x**	x	Yes	Yes	phase_a					Import Apparent Energy				
0x92	SenergyCT2	CT2 Apparent Energy	Energy Registers	GetImportApparentEnergy	**x**	x	Yes	Yes	phase_b					Import Apparent Energy				
0x93	SenergyCT3	CT3 Apparent Energy	Energy Registers	GetImportApparentEnergy	**x**	x	Yes	Yes	phase_c					Import Apparent Energy				
0x94	SVenergyT	Total Apparent Energy (Arit)	Energy Registers E36		**x**	x	No	Yes										
0x95	EnStatus0	Metering Status 0	Energy Registers E36	GetMeterStatus0	**0x0000**	0xF000	No	Yes						E0:0x0 or 0x5				
0x96	EnStatus1	Metering Status 1	Energy Registers E36	GetMeterStatus1	**0x0000**	0x0000	No	Yes						E1:0x0				
0x98	SVmeanT	Total Apparent Energy (Vect)	Energy Registers E36		**x**	x	No	Yes										
0x99	SVmeanTLSB	LSB of Vector Sum	Energy Registers E36		**x**	x	No	Yes										
0xA0	APenergyTF	Total Forward Fund. Energy	Fundamental Harmonic Energy Calibration Registers		**x**	x	Yes	Yes										
0xA1	APenergyCT1F	CT1 Forward Fund. Energy	Fundamental Harmonic Energy Calibration Registers		**x**	x	Yes	Yes	phase_a									
0xA2	APenergyCT2F	CT2 Forward Fund. Energy	Fundamental Harmonic Energy Calibration Registers		**x**	x	Yes	Yes	phase_b									
0xA3	APenergyCT3F	CT3 Forward Fund. Energy	Fundamental Harmonic Energy Calibration Registers		**x**	x	Yes	Yes	phase_c									
0xA4	ANenergyTF	Total Reverse Fund Energy	Fundamental Harmonic Energy Calibration Registers		**x**	x	Yes	Yes										
0xA5	ANenergyCT1F	CT1 Reverse Fund. Energy	Fundamental Harmonic Energy Calibration Registers		**x**	x	Yes	Yes	phase_a									

0xA6	ANenergyCT2F	CT2 Reverse Fund. Energy	Fundamental Harmonic Energy Calibration Registers		x	x	Yes	Yes	phase_b										
0xA7	ANenergyCT3F	CT3 Reverse Fund. Energy	Fundamental Harmonic Energy Calibration Registers		x	x	Yes	Yes	phase_c										
0xA8	APenergyTH	Total Forward Harm. Energy	Fundamental Harmonic Energy Calibration Registers		x	x	Yes	Yes											
0xA9	APenergyCT1H	CT1 Forward Harm. Energy	Fundamental Harmonic Energy Calibration Registers		x	x	Yes	Yes	phase_a										
0xAA	APenergyCT2H	CT2 Forward Harm. Energy	Fundamental Harmonic Energy Calibration Registers		x	x	Yes	Yes	phase_b										
0xAB	APenergyCT3H	CT3 Forward Harm. Energy	Fundamental Harmonic Energy Calibration Registers		x	x	Yes	Yes	phase_c										
0xAC	ANenergyTH	Total Reverse Harm. Energy	Fundamental Harmonic Energy Calibration Registers		x	x	Yes	Yes											
0xAD	ANenergyCT1H	CT1 Reverse Harm. Energy	Fundamental Harmonic Energy Calibration Registers		x	x	Yes	Yes	phase_a										
0xAE	ANenergyCT2H	CT2 Reverse Harm. Energy	Fundamental Harmonic Energy Calibration Registers		x	x	Yes	Yes	phase_b										
0xAF	ANenergyCT3H	CT3 Reverse Harm. Energy	Fundamental Harmonic Energy Calibration Registers		x	x	Yes	Yes	phase_c										
0xB0	PmeanT	Total Mean Power (P)	Power & Power Factor Registers		x	x	Yes	Yes											
0xB1	PmeanCT1	CT1 Mean Power (P)	Power & Power Factor Registers		x	x	Yes	Yes	phase_a										
0xB2	PmeanCT2	CT2 Mean Power (P)	Power & Power Factor Registers		x	x	Yes	Yes	phase_b										
0xB3	PmeanCT3	CT3 Mean Power (P)	Power & Power Factor Registers		x	x	Yes	Yes	phase_c										
0xB4	QmeanT	Total Mean Power (Q)	Power & Power Factor Registers		x	x	Yes	Yes											
0xB5	QmeanCT1	CT1 Mean Power (Q)	Power & Power Factor Registers	GetReactivePowerCT	x	x	Yes	Yes	phase_a	reactive_power				Value E32 val * 0.00032 E36 apower / 1000					
0xB6	QmeanCT2	CT2 Mean Power (Q)	Power & Power Factor Registers	GetReactivePowerCT	x	x	Yes	Yes	phase_b	reactive_power				Value E32 val * 0.00032 E36 apower / 1000					
0xB7	QmeanCT3	CT3 Mean Power (Q)	Power & Power Factor Registers	GetReactivePowerCT	x	x	Yes	Yes	phase_c	reactive_power				Value E32 val * 0.00032 E36 apower / 1000					
0xB8	SmeanT	Total Mean Power (S)	Power & Power Factor Registers		x	x	Yes	Yes											
0xB9	SmeanCT1	CT1 Mean Power (S)	Power & Power Factor Registers	GetApparentPowerCT	x	x	Yes	Yes	phase_a					32*1000					
0xBA	SmeanCT2	CT2 Mean Power (S)	Power & Power Factor Registers	GetApparentPowerCT	x	x	Yes	Yes	phase_b					32*1000					
0xBB	SmeanCT3	CT3 Mean Power (S)	Power & Power Factor Registers	GetApparentPowerCT	x	x	Yes	Yes	phase_c					32*1000					
0xBC	PFmeanT	Mean Power Factor	Power & Power Factor Registers		x	x	Yes	Yes											
0xBD	PFmeanCT1	CT1 Power Factor	Power & Power Factor Registers	GetPowerFactorCT	x	x	Yes	Yes	phase_a					32*1000					
0xBE	PFmeanCT2	CT2 Power Factor	Power & Power Factor Registers	GetPowerFactorCT	x	x	Yes	Yes	phase_b					32*1000					
0xBF	PFmeanCT3	CT3 Power Factor	Power & Power Factor Registers	GetPowerFactorCT	x	x	Yes	Yes	phase_c					32*1000					
0xC0	PmeanTLSB	Lower Word (Tot. Act. Power)	Power & Power Factor Registers		x	x	Yes	Yes											
0xC1	PmeanCT1LSB	Lower Word (CT1 Act. Power)	Power & Power Factor Registers		x	x	Yes	Yes	phase_a										
0xC2	PmeanCT2LSB	Lower Word (CT2 Act. Power)	Power & Power Factor Registers		x	x	Yes	Yes	phase_b										
0xC3	PmeanCT3LSB	Lower Word (CT3 Act. Power)	Power & Power Factor Registers		x	x	Yes	Yes	phase_c										
0xC4	QmeanTLSB	Lower Word (Tot. React. Power)	Power & Power Factor Registers		x	x	Yes	Yes											
0xC5	QmeanCT1LSB	Lower Word (CT1 React. Power)	Power & Power Factor Registers		x	x	Yes	Yes	phase_a										
0xC6	QmeanCT2LSB	Lower Word (CT2 React. Power)	Power & Power Factor Registers		x	x	Yes	Yes	phase_b										
0xC7	QmeanCT3LSB	Lower Word (CT3 React. Power)	Power & Power Factor Registers		x	x	Yes	Yes	phase_c										
0xC8	SAmeanTLSB	Lower Word (Tot. App. Power)	Power & Power Factor Registers		x	x	Yes	Yes											
0xC9	SmeanCT1LSB	Lower Word (CT1 App. Power)	Power & Power Factor Registers		x	x	Yes	Yes	phase_a										
0xCA	SmeanCT2LSB	Lower Word (CT2 App. Power)	Power & Power Factor Registers		x	x	Yes	Yes	phase_b										
0xCB	SmeanCT3LSB	Lower Word (CT3 App. Power)	Power & Power Factor Registers		x	x	Yes	Yes	phase_c										
0xD0	PmeanTF	Total Active Fund. Power	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes											
0xD1	PmeanCT1F	CT1 Active Fund. Power	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_a										
0xD2	PmeanCT2F	CT2 Active Fund. Power	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_b										
0xD3	PmeanCT3F	CT3 Active Fund. Power	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_c										
0xD4	PmeanTH	Total Active Harm. Power	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes											
0xD5	PmeanCT1H	CT1 Active Harm. Power	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_a										
0xD6	PmeanCT2H	CT2 Active Harm. Power	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_b										
0xD7	PmeanCT3H	CT3 Active Harm. Power	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_c										
0xD9	URMSCT1	CT1 RMS Voltage	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_a					Line 1 Voltage RMS					
0xDA	URMSCT2	CT2 RMS Voltage	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_b					Line 2 Voltage RMS					
0xDB	URMSCT3	CT3 RMS Voltage	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_c					Line 3 Voltage RMS					
0xDC	IRMSN	Calculated N RMS Current	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes											
0xDD	IRMSCT1	CT1 RMS Current	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_a					CT1 Current Clamp					
0xDE	IRMSCT2	CT2 RMS Current	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_b					CT2 Current Clamp					
0xDF	IRMSCT3	CT3 RMS Current	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_c					CT3 Current Clamp					
0xE0	PmeanTFLSB	Lower Word (Tot. Act. Fund. Power)	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes											
0xE1	PmeanCT1FLSB	Lower Word (CT1 Act. Fund. Power)	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_a										
0xE2	PmeanCT2FLSB	Lower Word (CT2 Act. Fund. Power)	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_b										
0xE3	PmeanCT3FLSB	Lower Word (CT3 Act. Fund. Power)	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_c										
0xE4	PmeanTHLSB	Lower Word (Tot. Act. Harm. Power)	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes											
0xE5	PmeanCT1HLSB	Lower Word (CT1 Act. Harm. Power)	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_a										
0xE6	PmeanCT2HLSB	Lower Word (CT2 Act. Harm. Power)	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_b										
0xE7	PmeanCT3HLSB	Lower Word (CT3 Act. Harm. Power)	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_c										
0xE9	URMSCT1LSB	Lower Word (CT1 RMS Voltage)	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_a										
0xEA	URMSCT2LSB	Lower Word (CT2 RMS Voltage)	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_b										
0xEB	URMSCT3LSB	Lower Word (CT3 RMS Voltage)	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_c										
0xED	IRMSCT1LSB	Lower Word (CT1 RMS Current)	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_a										
0xEE	IRMSCT2LSB	Lower Word (CT2 RMS Current)	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_b										
0xEF	IRMSCT3LSB	Lower Word (CT3 RMS Current)	Funamental & Harmonic Power & V/I RMS Registers		x	x	Yes	Yes	phase_c										
0xF1	THDNUCT1	CT1 Voltage THD+N	Peak Frequency Angle and Temperature Registers E36	GetVHarmCT	x	x	No	Yes	phase_a					Voltage Harmonics					
0xF2	THDNUCT2	CT2 Voltage THD+N	Peak Frequency Angle and Temperature Registers E36	GetVHarmCT	x	x	No	Yes	phase_b					Voltage Harmonics					
0xF3	THDNUCT3	CT3 Voltage THD+N	Peak Frequency Angle and Temperature Registers E36	GetVHarmCT	x	x	No	Yes	phase_c					Voltage Harmonics					
0xF4	Reserved	Reserved Register			x	x	Yes	Yes											
0xF5	THDNICT1	CT1 Current THD+N	Peak Frequency Angle and Temperature Registers E36	GetCHarmCT	x	x	No	Yes	phase_a					Current Harmonics					
0xF6	THDNICT2	CT2 Current THD+N	Peak Frequency Angle and Temperature Registers E36	GetCHarmCT	x	x	No	Yes	phase_b					Current Harmonics					
0xF7	THDNICT3	V Current THD+N	Peak Frequency Angle and Temperature Registers E36	GetCHarmCT	x	x	No	Yes	phase_c					Current Harmonics					
0xF8	Freq	Frequency	Peak Frequency Angle and Temperature Registers		x	x	Yes	Yes	frequency	freq_status	name		Frequency Status	Mains Frequncy 50Hz or 60Hz					
0xF9	PAngleCT1	CT1 Mean Phase Angle	Peak Frequency Angle and Temperature Registers	GetPhaseCT	x	x	Yes	Yes	phase_a	phase_angle	name			Mean Phase Angle (angleA / 10)					
0xFA	PAngleCT2	CT2 Mean Phase Angle	Peak Frequency Angle and Temperature Registers	GetPhaseCT	x	x	Yes	Yes	phase_b	phase_angle	name			Mean Phase Angle (angleA / 10)					
0xFB	PAngleCT3	CT3 Mean Phase Angle	Peak Frequency Angle and Temperature Registers	GetPhaseCT	x	x	Yes	Yes	phase_c	phase_angle	name			Mean Phase Angle (angleA / 10)					
0xFC	Temp	Measured Temperature	Peak Frequency Angle and Temperature Registers	GetTemperature	x	x	Yes	Yes	chip_temperature										
0xFD	UangleCT1	CT1 Voltage Phase Angle	Peak Frequency Angle and Temperature Registers		x	x	Yes	Yes	phase_a					Phase A Voltage Phase Angle					
0xFE	UangleCT2	CT2 Voltage Phase Angle	Peak Frequency Angle and Temperature Registers		x	x	Yes	Yes	phase_b					Phase B Voltage Phase Angle					
0xFF	UangleCT3	CT3 Voltage Phase Angle	Peak Frequency Angle and Temperature Registers		x	x	Yes	Yes	phase_c					Phase C Voltage Phase Angle					
0x100	AI_HR2	phase A, Current, Harmonic Ratio for 2-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x101	AI_HR3	phase A, Current, Harmonic Ratio for 3-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x102	AI_HR4	phase A, Current, Harmonic Ratio for 4-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x103	AI_HR5	phase A, Current, Harmonic Ratio for 5-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x104	AI_HR6	phase A, Current, Harmonic Ratio for 6-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x105	AI_HR7	phase A, Current, Harmonic Ratio for 7-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x106	AI_HR8	phase A, Current, Harmonic Ratio for 8-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x107	AI_HR9	phase A, Current, Harmonic Ratio for 9-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x108	AI_HR10	phase A, Current, Harmonic Ratio for 10-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x109	AI_HR11	phase A, Current, Harmonic Ratio for 11-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x10A	AI_HR12	phase A, Current, Harmonic Ratio for 12-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x10B	AI_HR13	phase A, Current, Harmonic Ratio for 13-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x10C	AI_HR14	phase A, Current, Harmonic Ratio for 14-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x10D	AI_HR15	phase A, Current, Harmonic Ratio for 15-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x10E	AI_HR16	phase A, Current, Harmonic Ratio for 16-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x10F	AI_HR17	phase A, Current, Harmonic Ratio for 17-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x110	AI_HR18	phase A, Current, Harmonic Ratio for 18-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x111	AI_HR19	phase A, Current, Harmonic Ratio for 19-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x112	AI_HR20	phase A, Current, Harmonic Ratio for 20-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x113	AI_HR21	phase A, Current, Harmonic Ratio for 21-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x114	AI_HR22	phase A, Current, Harmonic Ratio for 22-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x115	AI_HR23	phase A, Current, Harmonic Ratio for 23-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x116	AI_HR24	phase A, Current, Harmonic Ratio for 24-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x117	AI_HR25	phase A, Current, Harmonic Ratio for 25-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x118	AI_HR26	phase A, Current, Harmonic Ratio for 26-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x119	AI_HR27	phase A, Current, Harmonic Ratio for 27-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x11A	AI_HR28	phase A, Current, Harmonic Ratio for 28-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x11B	AI_HR29	phase A, Current, Harmonic Ratio for 29-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x11C	AI_HR30	phase A, Current, Harmonic Ratio for 30-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x11D	AI_HR31	phase A, Current, Harmonic Ratio for 31-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x11E	AI_HR32	phase A, Current, Harmonic Ratio for 32-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x11F	AI_THD	phase A, Current, Total Harmonic Distortion Ratio	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a										
0x120	BI_HR2	phase B, Current, Harmonic Ratio for 2-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b										

0x121	BI_HR3	phase B, Current, Harmonic Ratio for 3-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x122	BI_HR4	phase B, Current, Harmonic Ratio for 4-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x123	BI_HR5	phase B, Current, Harmonic Ratio for 5-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x124	BI_HR6	phase B, Current, Harmonic Ratio for 6-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x125	BI_HR7	phase B, Current, Harmonic Ratio for 7-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x126	BI_HR8	phase B, Current, Harmonic Ratio for 8-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x127	BI_HR9	phase B, Current, Harmonic Ratio for 9-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x128	BI_HR10	phase B, Current, Harmonic Ratio for 10-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x129	BI_HR11	phase B, Current, Harmonic Ratio for 11-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x12A	BI_HR12	phase B, Current, Harmonic Ratio for 12-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x12B	BI_HR13	phase B, Current, Harmonic Ratio for 13-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x12C	BI_HR14	phase B, Current, Harmonic Ratio for 14-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x12D	BI_HR15	phase B, Current, Harmonic Ratio for 15-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x12E	BI_HR16	phase B, Current, Harmonic Ratio for 16-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x12F	BI_HR17	phase B, Current, Harmonic Ratio for 17-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x130	BI_HR18	phase B, Current, Harmonic Ratio for 18-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x131	BI_HR19	phase B, Current, Harmonic Ratio for 19-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x132	BI_HR20	phase B, Current, Harmonic Ratio for 20-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x133	BI_HR21	phase B, Current, Harmonic Ratio for 21-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x134	BI_HR22	phase B, Current, Harmonic Ratio for 22-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x135	BI_HR23	phase B, Current, Harmonic Ratio for 23-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x136	BI_HR24	phase B, Current, Harmonic Ratio for 24-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x137	BI_HR25	phase B, Current, Harmonic Ratio for 25-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x138	BI_HR26	phase B, Current, Harmonic Ratio for 26-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x139	BI_HR27	phase B, Current, Harmonic Ratio for 27-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x13A	BI_HR28	phase B, Current, Harmonic Ratio for 28-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x13B	BI_HR29	phase B, Current, Harmonic Ratio for 29-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x13C	BI_HR30	phase B, Current, Harmonic Ratio for 30-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x13D	BI_HR31	phase B, Current, Harmonic Ratio for 31-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x13E	BI_HR32	phase B, Current, Harmonic Ratio for 32-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x13F	BI_THD	phase B, Current, Total Harmonic Distortion Ratio	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x140	CI_HR2	phase C, Current, Harmonic Ratio for 2-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x141	CI_HR3	phase C, Current, Harmonic Ratio for 3-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x142	CI_HR4	phase C, Current, Harmonic Ratio for 4-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x143	CI_HR5	phase C, Current, Harmonic Ratio for 5-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x144	CI_HR6	phase C, Current, Harmonic Ratio for 6-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x145	CI_HR7	phase C, Current, Harmonic Ratio for 7-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x146	CI_HR8	phase C, Current, Harmonic Ratio for 8-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x147	CI_HR9	phase C, Current, Harmonic Ratio for 9-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x148	CI_HR10	phase C, Current, Harmonic Ratio for 10-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x149	CI_HR11	phase C, Current, Harmonic Ratio for 11-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x14A	CI_HR12	phase C, Current, Harmonic Ratio for 12-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x14B	CI_HR13	phase C, Current, Harmonic Ratio for 13-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x14C	CI_HR14	phase C, Current, Harmonic Ratio for 14-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x14D	CI_HR15	phase C, Current, Harmonic Ratio for 15-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x14E	CI_HR16	phase C, Current, Harmonic Ratio for 16-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x14F	CI_HR17	phase C, Current, Harmonic Ratio for 17-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x150	CI_HR18	phase C, Current, Harmonic Ratio for 18-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x151	CI_HR19	phase C, Current, Harmonic Ratio for 19-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x152	CI_HR20	phase C, Current, Harmonic Ratio for 20-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x153	CI_HR21	phase C, Current, Harmonic Ratio for 21-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x154	CI_HR22	phase C, Current, Harmonic Ratio for 22-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x155	CI_HR23	phase C, Current, Harmonic Ratio for 23-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x156	CI_HR24	phase C, Current, Harmonic Ratio for 24-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x157	CI_HR25	phase C, Current, Harmonic Ratio for 25-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x158	CI_HR26	phase C, Current, Harmonic Ratio for 26-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x159	CI_HR27	phase C, Current, Harmonic Ratio for 27-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x15A	CI_HR28	phase C, Current, Harmonic Ratio for 28-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x15B	CI_HR29	phase C, Current, Harmonic Ratio for 29-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x15C	CI_HR30	phase C, Current, Harmonic Ratio for 30-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x15D	CI_HR31	phase C, Current, Harmonic Ratio for 31-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x15E	CI_HR32	phase C, Current, Harmonic Ratio for 32-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x15F	CI_THD	phase C, Current, Total Harmonic Distortion Ratio	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x160	AV_HR2	phase A, Voltage, Harmonic Ratio for 2-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x161	AV_HR3	phase A, Voltage, Harmonic Ratio for 3-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x162	AV_HR4	phase A, Voltage, Harmonic Ratio for 4-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x163	AV_HR5	phase A, Voltage, Harmonic Ratio for 5-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x164	AV_HR6	phase A, Voltage, Harmonic Ratio for 6-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x165	AV_HR7	phase A, Voltage, Harmonic Ratio for 7-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x166	AV_HR8	phase A, Voltage, Harmonic Ratio for 8-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x167	AV_HR9	phase A, Voltage, Harmonic Ratio for 9-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x168	AV_HR10	phase A, Voltage, Harmonic Ratio for 10-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x169	AV_HR11	phase A, Voltage, Harmonic Ratio for 11-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x16A	AV_HR12	phase A, Voltage, Harmonic Ratio for 12-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x16B	AV_HR13	phase A, Voltage, Harmonic Ratio for 13-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x16C	AV_HR14	phase A, Voltage, Harmonic Ratio for 14-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x16D	AV_HR15	phase A, Voltage, Harmonic Ratio for 15-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x16E	AV_HR16	phase A, Voltage, Harmonic Ratio for 16-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x16F	AV_HR17	phase A, Voltage, Harmonic Ratio for 17-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x170	AV_HR18	phase A, Voltage, Harmonic Ratio for 18-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x171	AV_HR19	phase A, Voltage, Harmonic Ratio for 19-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x172	AV_HR20	phase A, Voltage, Harmonic Ratio for 20-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x173	AV_HR21	phase A, Voltage, Harmonic Ratio for 21-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x174	AV_HR22	phase A, Voltage, Harmonic Ratio for 22-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x175	AV_HR23	phase A, Voltage, Harmonic Ratio for 23-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x176	AV_HR24	phase A, Voltage, Harmonic Ratio for 24-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x177	AV_HR25	phase A, Voltage, Harmonic Ratio for 25-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x178	AV_HR26	phase A, Voltage, Harmonic Ratio for 26-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x179	AV_HR27	phase A, Voltage, Harmonic Ratio for 27-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x17A	AV_HR28	phase A, Voltage, Harmonic Ratio for 28-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x17B	AV_HR29	phase A, Voltage, Harmonic Ratio for 29-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x17C	AV_HR30	phase A, Voltage, Harmonic Ratio for 30-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x17D	AV_HR31	phase A, Voltage, Harmonic Ratio for 31-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x17E	AV_HR32	phase A, Voltage, Harmonic Ratio for 32-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x17F	AV_THD	phase A, Voltage, Total Harmonic Distortion Ratio	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x180	BV_HR2	phase B, Voltage, Harmonic Ratio for 2-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x181	BV_HR3	phase B, Voltage, Harmonic Ratio for 3-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x182	BV_HR4	phase B, Voltage, Harmonic Ratio for 4-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x183	BV_HR5	phase B, Voltage, Harmonic Ratio for 5-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x184	BV_HR6	phase B, Voltage, Harmonic Ratio for 6-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x185	BV_HR7	phase B, Voltage, Harmonic Ratio for 7-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x186	BV_HR8	phase B, Voltage, Harmonic Ratio for 8-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x187	BV_HR9	phase B, Voltage, Harmonic Ratio for 9-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x188	BV_HR10	phase B, Voltage, Harmonic Ratio for 10-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x189	BV_HR11	phase B, Voltage, Harmonic Ratio for 11-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x18A	BV_HR12	phase B, Voltage, Harmonic Ratio for 12-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x18B	BV_HR13	phase B, Voltage, Harmonic Ratio for 13-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x18C	BV_HR14	phase B, Voltage, Harmonic Ratio for 14-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x18D	BV_HR15	phase B, Voltage, Harmonic Ratio for 15-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x18E	BV_HR16	phase B, Voltage, Harmonic Ratio for 16-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x18F	BV_HR17	phase B, Voltage, Harmonic Ratio for 17-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x190	BV_HR18	phase B, Voltage, Harmonic Ratio for 18-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x191	BV_HR19	phase B, Voltage, Harmonic Ratio for 19-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x192	BV_HR20	phase B, Voltage, Harmonic Ratio for 20-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x193	BV_HR21	phase B, Voltage, Harmonic Ratio for 21-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									

0x194	BV_HR22	phase B, Voltage, Harmonic Ratio for 22-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x195	BV_HR23	phase B, Voltage, Harmonic Ratio for 23-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x196	BV_HR24	phase B, Voltage, Harmonic Ratio for 24-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x197	BV_HR25	phase B, Voltage, Harmonic Ratio for 25-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x198	BV_HR26	phase B, Voltage, Harmonic Ratio for 26-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x199	BV_HR27	phase B, Voltage, Harmonic Ratio for 27-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x19A	BV_HR28	phase B, Voltage, Harmonic Ratio for 28-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x19B	BV_HR29	phase B, Voltage, Harmonic Ratio for 29-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x19C	BV_HR30	phase B, Voltage, Harmonic Ratio for 30-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x19D	BV_HR31	phase B, Voltage, Harmonic Ratio for 31-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x19E	BV_HR32	phase B, Voltage, Harmonic Ratio for 32-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x19F	BV_THD	phase B, Voltage, Total Harmonic Distortion Ratio	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x1A0	CV_HR2	phase C, Voltage, Harmonic Ratio for 2-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1A1	CV_HR3	phase C, Voltage, Harmonic Ratio for 3-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1A2	CV_HR4	phase C, Voltage, Harmonic Ratio for 4-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1A3	CV_HR5	phase C, Voltage, Harmonic Ratio for 5-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1A4	CV_HR6	phase C, Voltage, Harmonic Ratio for 6-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1A5	CV_HR7	phase C, Voltage, Harmonic Ratio for 7-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1A6	CV_HR8	phase C, Voltage, Harmonic Ratio for 8-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1A7	CV_HR9	phase C, Voltage, Harmonic Ratio for 9-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1A8	CV_HR10	phase C, Voltage, Harmonic Ratio for 10-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1A9	CV_HR11	phase C, Voltage, Harmonic Ratio for 11-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1AA	CV_HR12	phase C, Voltage, Harmonic Ratio for 12-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1AB	CV_HR13	phase C, Voltage, Harmonic Ratio for 13-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1AC	CV_HR14	phase C, Voltage, Harmonic Ratio for 14-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1AD	CV_HR15	phase C, Voltage, Harmonic Ratio for 15-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1AE	CV_HR16	phase C, Voltage, Harmonic Ratio for 16-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1AF	CV_HR17	phase C, Voltage, Harmonic Ratio for 17-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1B0	CV_HR18	phase C, Voltage, Harmonic Ratio for 18-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1B1	CV_HR19	phase C, Voltage, Harmonic Ratio for 19-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1B2	CV_HR20	phase C, Voltage, Harmonic Ratio for 20-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1B3	CV_HR21	phase C, Voltage, Harmonic Ratio for 21-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1B4	CV_HR22	phase C, Voltage, Harmonic Ratio for 22-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1B5	CV_HR23	phase C, Voltage, Harmonic Ratio for 23-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1B6	CV_HR24	phase C, Voltage, Harmonic Ratio for 24-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1B7	CV_HR25	phase C, Voltage, Harmonic Ratio for 25-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1B8	CV_HR26	phase C, Voltage, Harmonic Ratio for 26-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1B9	CV_HR27	phase C, Voltage, Harmonic Ratio for 27-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1BA	CV_HR28	phase C, Voltage, Harmonic Ratio for 28-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1BB	CV_HR29	phase C, Voltage, Harmonic Ratio for 29-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1BC	CV_HR30	phase C, Voltage, Harmonic Ratio for 30-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1BD	CV_HR31	phase C, Voltage, Harmonic Ratio for 31-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1BE	CV_HR32	phase C, Voltage, Harmonic Ratio for 32-th order component	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1BF	CV_THD	phase C, Voltage, Total Harmonic Distortion Ratio	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1C0	AI_FUND	phase A, Current, Fundamental component value	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x1C1	AV_FUND	phase A, Voltage, Fundamental component value	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_a									
0x1C2	BI_FUND	Phase B, Current, Fundamental component value	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x1C3	BV_FUND	Phase B, Voltage, Fundamental component value	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_b									
0x1C4	CI_FUND	Phase C, Current, Fundamental component value	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1C5	CV_FUND	Phase C, Voltage, Fundamental component value	Harmonic Fourier Analysis Results Registers		x	x	No	Yes	phase_c									
0x1D0	DFT_SCALE	DFT Scale Configuration	Harmonic Fourier Analysis Results Registers		x	x	No	Yes										
0x1D1	DFT_CTRL	DFT Control Configuration	Harmonic Fourier Analysis Results Registers		x	x	No	Yes										